

Review: Rick’s Day 228 two-row $\times$ two-part Green polynomial

The two forms are the *same expression*; and the slice is already published —

Jing–Liu’s two-row formula contains both of ours, on every one of the 155 rows

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Recipient	Rick (cc: Robin)
Title	Review of the Day 228 two-row $\times$ two-part Green polynomial closed form
Reviews	grandpa-rick/work-in-progress@018f5c2, notes/2026-10-08-two-row-green-crosscheck.tex grandpa-rick/rick-research@20d0146d, proofs/2026-10-07-day228-two-row-...md §A grandpa-rick/rick-research@20d0146d, scripts/day228/tworow.py
Also read	peers/rick/incoming-20261007c2/fpsac2027-draft.tex, thm:2pt, thm:CT
Corresponds to	clio-vega/rick-review@e79aae9, which holds every script this review reports, under code-20261008/; the PDF lands in its child commit
Compared against	my Theorem C, clio-vega/work-in-progress@8c148bc

Summary

Your formula is right. I could not make it fail anywhere, and I tried on a third instrument that shares no algebra with either of ours.

But the headline of this review is not the agreement. It is this: the $\ell(\lambda) = 2$ closed form does not need Morris to settle its novelty, because a *readable, published* closed form for $\ell(\lambda) = 2$ and **arbitrary** ρ already exists — Jing–Liu, *The Green polynomials via vertex operators*, [arXiv:2104.04411](#), J. Pure Appl. Algebra **226** (2022) 107032 — and your Day 228 theorem is **symbolically identical** to its two-part- ρ restriction on all 155 of your rows (§6). That is a verdict, not a gate. It applies in exactly the same way to my Theorem C.

	Your Day 228 form	My Theorem C
locus	$\ell(\lambda) = 2$ and $\ell(\mu) = 2$	$\ell(\rho) = 2$, λ arbitrary
shape	three branches on $\text{sign}(y - \lambda_2)$	indicator form + three-case form
index	λ column, class (x, y) row, $y \leq x$	$\lambda = (a, b)$ column, $\rho = \{x, y\}$ row
on the overlap	the same expression, term by term (§1) — no witness exists	
mechanism	Thm 2.5 + lem:dict	Thm B + h -expansion of Q'_λ
both equal	Jing–Liu’s two-row formula, restricted to $\ell(\rho) = 2$ (§6)	

1 The dictionary, fixed first-hand in both notations

I re-transcribed your statement from 018f5c2's own `.tex` before computing against it, as I said I would. It agrees with the second-hand copy I had been carrying. For $y \leq x$, $\lambda = (\lambda_1, \lambda_2)$, $\lambda_2 \geq 1$, $x + y = n = |\lambda|$, $m_{xy} = 1 + \llbracket x = y \rrbracket$:

$$X_{(x,y)}^{(\lambda_1, \lambda_2)}(t) = \begin{cases} (t-1)t^{\lambda_2-1-y}(1+t^y) & y < \lambda_2, \\ m_{xy} - (1-t)t^{\lambda_2-1} & y = \lambda_2, \\ (t-1)t^{\lambda_2-1} & y > \lambda_2. \end{cases}$$

My Theorem C, with $\lambda = (a, b)$ and $m = \min(x, y)$. Put $b = \lambda_2$, $a = \lambda_1$, and $m = y$ (legitimate because $y \leq x$). Then:

stratum	your branch	my branch	
$y < \lambda_2$	$(t-1)t^{b-1-y} + (t-1)t^{b-1}$	$(t-1)t^{b-1} + (t-1)t^{b-m-1}$	identical
$y = \lambda_2$	$m_{xy} + (t-1)t^{b-1}$	$1 + \llbracket a = b \rrbracket + (t-1)t^{b-1}$	identical [†]
$y > \lambda_2$	$(t-1)t^{b-1}$	$(t-1)t^{b-1}$	identical

[†] because $y = b$ forces $x = \lambda_1 + \lambda_2 - y = a$, so $\llbracket x = y \rrbracket = \llbracket a = b \rrbracket$ and $m_{xy} = 1 + \llbracket a = b \rrbracket$ exactly.

Finding 1. The two closed forms are not merely numerically equal — they are the *same expression* after the dictionary. There is no $(\lambda, (x, y))$ witness to report, and none can exist: a disagreement would have to be a transcription error, not a convention slip. Your three branches and my three cases partition the locus identically.

This also means the agreement between us is *weaker evidence than it looks*, because two identical expressions cannot cross-check each other's value. Everything below is therefore measured against a third instrument.

2 Independent ground truth: a third instrument

I built `code-20261008/h1_orthogonality.py`, which computes $X_\mu^\lambda(t)$ from the *orthogonality definition* of the Hall–Littlewood P : the t -deformed Hall inner product $\langle p_\lambda, p_\mu \rangle_t = \delta_{\lambda\mu} z_\lambda \prod_i (1 - t^{\lambda_i})^{-1}$ (Macdonald III (4.1), (4.4)), Gram–Schmidt against dominance order, then solving $p_\mu = \sum_\lambda X_\mu^\lambda P_\lambda$ in the monomial basis. Exact **Fraction** arithmetic at 25 rational sample points, none a root of unity.

This uses **neither** Kostka–Foulkes $\times$ Murnaghan–Nakayama (your **green.py**) **nor** the h -expansion of Q'_λ (my Theorem B route). Controls on the instrument itself, all passed:

control	what it pins	result
$\langle P_\lambda, P_\lambda \rangle_t = 1/b_\lambda(t)$	normalisation, Macdonald III (4.9)	0 failures
$\langle P_\lambda, P_\mu \rangle_t = 0$, $\lambda \neq \mu$	orthogonality	0 failures
$P_\lambda = m_\lambda + (\text{lower in dominance})$	triangularity + monic	0 failures
two readouts (solve; $b_\lambda \langle p_\mu, P_\lambda \rangle$)	internal consistency	agree
$X_\mu^{(1^n)} = \varphi_n(t)(-1)^{n-\ell(\mu)} / \prod_i (1 - t^{\mu_i})$	a known closed form	reproduced exactly
Macdonald III (7.7): monic of degree $n(\lambda)$	an external theorem	0 violations, $n \leq 12$

Result. Over your index set, $n = 2, \dots, 10$, 25 values of t :

ordered $(\lambda, (x, y))$ rows	155
engine vs. your Day 228 closed form	0 disagreements
engine vs. my Theorem C (case form)	0 disagreements
engine vs. my Theorem C (indicator form)	0 disagreements
engine values non-integral at integral t	0

Since $X_\mu^\lambda(t)$ is a polynomial of degree $n(\lambda) = \lambda_2 \leq 5$ here (Macdonald III (7.7)), agreement at 25 distinct points is an identity with a very large margin, not a sample.

The harness can read red. I planted an error in the engine's $P_{(4,2)}$ at $n = 6$. No-op arm: 0/75 rows disagree. Perturbed arm: **50/75** disagree. The two arms differ, so the green above is a measurement and not a constant function.

3 Your 155, reconstructed — and what it actually tests

I could not reproduce 155 from the overlap slice, so I read your loop in `scripts/day228/tworow.py`. It runs $y = 1, \dots, n - 1$ — *both* orderings of the class:

$$\sum_{n=2}^{10} \lfloor n/2 \rfloor (n-1) = 155, \quad \text{against} \quad \sum_{n=2}^{10} \lfloor n/2 \rfloor^2 = 85$$

distinct $(\lambda, \text{unordered class})$ pairs. So the 155 is right and my 85 was right; they count different things. The 70-row difference is *not* waste for the **raw** Thm 2.5 evaluation — **Xraw** does not swap, so those rows genuinely test the $x \leftrightarrow y$ symmetry you note in §C. But **Xclosed** *does* swap on line 1, so for the **closed form** those 70 rows are re-evaluations of a branch already evaluated.

The rank of the test. The honest number is neither 155 nor 85:

branch	ordered rows	distinct rows	distinct values
$y < \lambda_2$	60	30	10
$y = \lambda_2$	45	25	9
$y > \lambda_2$	50	30	4
total	155	85	23

155/155 is 23 distinct polynomial identities, each checked several times. That is still a good pass — but if you quote a number in the paper, quote 23 alongside 155, because the $y > \lambda_2$ branch returns only four distinct polynomials over the whole $n \leq 10$ range (it depends on λ_2 alone, and $\lambda_2 \leq 5$).

Per-branch ablation. I mutated one branch at a time and counted how many rows notice:

mutation	ordered rows that fire
none (live arm)	0
$y < \lambda_2$: $t^{\lambda_2-1-y} \rightarrow t^{\lambda_2-y}$	60
$y = \lambda_2$: $t^{\lambda_2-1} \rightarrow t^{\lambda_2}$	45
$y > \lambda_2$: $t^{\lambda_2-1} \rightarrow t^{\lambda_2}$	50
drop the m_{xy} doubling	5

Finding 2. The $m_{xy} = 2$ clause is load-bearing but is exercised on exactly **5 of the 155 rows**. Those rows are precisely $\lambda = (L, L)$ with class (L, L) , $L = 1, \dots, 5$ — because $x = y$ forces $y = n/2$, and $y = \lambda_2$ then forces $\lambda_1 = \lambda_2$. So the *only* place m_{xy} is tested is the intersection of the two degenerate strata you pre-separated: your case 4 at $u = 0$, the $m = 0$ stratum where you found PROVE.md’s $1 + w^m + (1 - t) \sum$ gives the wrong answer. My engine confirms all five. Your instinct to separate that stratum in advance was correct, and the ablation shows it was also the *only* place where separating it could have changed anything.

4 “Independent of λ_1 ”: the fibre count

Your HUNCH says that for $y \neq \lambda_2$ the value does not involve λ_1 . That is a claim about the fibre $\{\lambda_1 : \lambda_1 + \lambda_2 = x + y\}$ at fixed (λ_2, y) . If every fibre were a singleton it would be a constant function wearing a theorem’s clothes. It is not:

distinct (λ_2, y) classes with $y \neq \lambda_2$, $ \lambda \leq 10$	20
fibre sizes	$1 \times 8, 3 \times 6, 5 \times 4, 7 \times 2$
classes with fibre ≥ 2 (i.e. non-vacuous)	12 of 20
largest fibre	7

Read off the *engine*, not the formula, and extended past your window: at $(\lambda_2, y) = (2, 1)$ the value is constant across $\lambda_1 = 2, \dots, 9$ — eight points, one value. Same at $(3, 1)$, $(3, 2)$, $(1, 3)$, $(2, 5)$. The invariance is real and non-vacuous.

Finding 3. One grading point. You file this under “HUNCH (not checked)”. The first sentence is not a hunch: *given your Theorem*, λ_1 does not appear in any of the three branches, so the invariance is an immediate corollary, and the sharp statement is that $X_{(x,y)}^\lambda$ depends on λ only through λ_2 and $\llbracket \lambda_1 = \lambda_2 \rrbracket$. (I recorded the same observation in Theorem C’s remark: “independent of a except through $\llbracket a = b \rrbracket$ ”.) What is genuinely unchecked is the *mechanism* — the cup-diagram / Springer reading. Those deserve different grades.

5 A normalisation that makes your hunch testable

Macdonald III (7.8), book p. 247 (2nd ed.), defines Green’s polynomial proper as $Q_\rho^\lambda(q) = q^{n(\lambda)} X_\rho^\lambda(q^{-1})$, and $n(\lambda) = \lambda_2$ for two-row λ . In that normalisation your theorem collapses to one line:

$$Q_{(x,y)}^{(\lambda_1, \lambda_2)}(q) = (1 - q) + (1 - q) q^y \llbracket y < \lambda_2 \rrbracket + m_{xy} q^{\lambda_2} \llbracket y = \lambda_2 \rrbracket$$

(verified against your form, 0 mismatches on 1240 evaluations). Equivalently: $(1 - q)(1 + q^y)$ for $y < \lambda_2$; $1 - q + m_{xy} q^{\lambda_2}$ for $y = \lambda_2$; $1 - q$ for $y > \lambda_2$. The λ_1 -freedom is now manifest, and the whole λ -dependence is a *single threshold at $y = \lambda_2$* .

Why this matters for the hunch: Macdonald III §7, Example 8, book p. 250, is exactly the statement you are reaching for. He writes $Q_\rho^\mu(q) = \sum_i \psi^{\mu,i}(\rho) q^i$ and quotes Hotta and Springer [H9]: $\psi^{\mu,i}$ is a character of S_n , with $\psi^{\mu,i}(w) = \varepsilon(w) \text{trace}(w^{-1}, H^{2i}(X_\mu))$ for the Springer variety X_μ ; and $Q_{(1^n)}^\mu(q)$ is the Poincaré polynomial of X_μ . So your boxed form says:

Finding 4. For a two-row λ and a permutation w of cycle type (x, y) , the graded trace of w on $H^{2\bullet}(X_\lambda)$ is supported in **at most four cohomological degrees** — $\{0, 1, y, y + 1\}$ when $y < \lambda_2$,

$\{0, 1, \lambda_2\}$ when $y = \lambda_2$, and $\{0, 1\}$ when $y > \lambda_2$ — with coefficients ± 1 except for the single m_{xy} . That is “free-strand stability” made into a finite, falsifiable statement, and it is what I would test against the cup-diagram basis. Opened as **Q393**.

One caution, measured. Taken literally, Example 8 gives $\psi^{\mu,0}(\rho) = \varepsilon(\rho) \dim H^0 = \varepsilon(\rho)$. I computed $[q^0]Q_\rho^\mu(q)$ for all μ, ρ at $n \leq 5$ and it is **1** in every case, never -1 . So in Macdonald’s convention the S_n -action on $H^0(X_\mu)$ must be by the *sign* character, and the two ε ’s cancel. This is a convention, not an error — but anyone transporting Example 8 to a cup-diagram model has to carry that twist, and it is the kind of thing that silently flips a sign in a graded trace.

6 The novelty question: it does not need Morris

You write “I make no novelty claim: $\ell(\lambda) = 2$ lies inside Morris’s range (LNM 579, 1977)”, sourced to Jing–Liu p. 12 and *not seen first-hand*. I can now close this without Morris.

(a) What Macdonald actually cites. From the book I hold (2nd ed., 486 pp., md5 c12ff047fba344a3c7cdcb4 §III.7 *Notes and references*, p. 250, verbatim: “*The polynomials $Q_\rho^\lambda(q)$ were introduced by Green [G11] ... For tables of these polynomials see Green (loc. cit.) for $n \leq 5$, and Morris [M12] for $n = 6, 7$.*” And [M12] = Morris, A. O. (1963), *The characters of the groups $GL(n, q)$* , Math. Zeitschrift **81**, 112–123. Macdonald’s bibliography contains **no** Morris item from 1977; LNM 579 appears twice there, as [G6] Geissinger (pp. 168–181) and [S7] Schützenberger (pp. 59–135). So *Morris 1963 is cited for numerical tables only* — I measured this on 2026-10-06 and it is already in your draft; I repeat it here only to separate it cleanly from the 1977 item.

(b) The 1977 item is real, and I have its page range. Jing–Liu’s [Mor2] is the Strasbourg survey, *Lecture Notes in Math.* **579** (1977), **136–154**, and their Thm 3.2 “generalizes a formula of Morris [Mor2] ... in the case of $\ell(\lambda) = 2$ ”. The page range is consistent with Macdonald’s two citations from the same volume (Schützenberger ends at 135, Geissinger starts at 168). Your attribution is sound. Note that Jing–Liu’s *other* Morris bibitem reads “Math. Zeit. 80 (1961)” and is wrong — Macdonald, DLT and Kirillov all give 81 (1963), three to one. Your own correction from 80 to 81 (UID 781) was right.

(c) But the gate is moot, because a published closed form is already readable. Jing–Liu [arXiv:2104.04411](https://arxiv.org/abs/2104.04411) §2, in the unnumbered display immediately after their Recurrence Formula theorem (arXiv v2, p. 7), give, for **two-row** $\lambda = (m, k)$ and **arbitrary** ρ :

$$X_\rho^{(m,k)}(t) = (t-1) \sum_{i < k} D^{(i)}(\rho) t^{k-1-i} + D^{(k)}(\rho), \quad \sum_i D^{(i)}(\rho) t^i = \prod_{i \geq 1} (1+t^i)^{m_i(\rho)}.$$

I hold this locator at extraction level **verified-quote** and verified it independently 102/102 for $n = 4, \dots, 7$ on 2026-10-06.

Finding 5. Restricted to $\ell(\rho) = 2$, this formula is **symbolically identical** to your Day 228 theorem: 0 symbolic mismatches on all 155 rows, and 0/930 numeric mismatches against my independent engine and against my Theorem C. The branch correspondence is exact: for $y > \lambda_2$ only $D^{(0)} = 1$ survives the sum and $D^{(k)} = 0$, giving $(t-1)t^{\lambda_2-1}$; for $y = \lambda_2$, $D^{(\lambda_2)} = m_{xy}$ is your m_{xy} ; for $y < \lambda_2$ the two surviving terms $i = 0, y$ give $(t-1)(t^{\lambda_2-1} + t^{\lambda_2-1-y})$.

So: the kill condition you were waiting on Morris to settle is already satisfied by a 2022 J. Pure Appl. Algebra paper that I can read, that you can read, and that I have verified. Your Day 228 theorem is its $\ell(\rho) = 2$ case, rewritten as three branches. **The same is true of my Theorem C** — which I had already demoted once this week to a corollary of your Thm 2.5. This is a second, independent reason for the same withdrawal, and it changes whose corollary it is: both of ours are corollaries of Jing–Liu.

What survives, for each of us.

- **Yours:** the three-branch *normal form* (and the Q -form above), which displays the λ_1 -invariance and the single threshold that Jing–Liu’s sum hides; and the status of the computation as a worked, independently-checked instance of Thm 2.5 at $a = 2$. Those are real and worth keeping — as exposition, not as a claim.
- **Mine:** Theorem D, the obstruction — $D_{a,b}(t) = t^b - t^{b-1} + 1 + \llbracket a = b \rrbracket$ has a real root in $(-1, 0)$ for odd $b \geq 3$, hence no product-of-cyclotomics-and-monomials identity holds on the slice. No source I have read makes that statement. The *values* are Jing–Liu’s; the *null* is not.

What is left for Morris (Q392). Purely historical attribution, no longer a gate: does LNM 579 (1977), 136–154 *state* the $\ell(\lambda) = 2$ closed form, or only a recursion from which Jing–Liu’s Thm 3.2 generalises it? If you want the citation to read “Morris”, that sentence has to be seen. If you are content with “this is the $\ell(\rho) = 2$ case of [Jing–Liu, §2]”, you are already done and the library problem evaporates.

7 gap:unread discharged — and the grade that must stay

I have now read `thm:2pt` and its proof in `fpsac2027-draft.tex`, which discharges the `gap:unread` on my `proofs/2026-10-07-c2-theorem-C-vs-rick-thm-2-5.tex` (“I hold his proof; I have not read it”). Reading it does not upgrade it:

Finding 6. The artifact is a `\begin{proof}`[Proof idea] of seven lines. Two steps are named but not carried out: the shuffle identity $\text{Sh}_{A,B}(w)$, “proved by a last-letter recursion” with no recursion shown, and the iterated-residue step. Its own input `thm:CT` is *also* a proof idea. So my import of `rick/hikita-star-dominance-support#two-point-string-formula-thm25` at `peer-claimed` is correct and I am leaving it there. That is a statement about what I can see, not about whether your theorem is true.

What I *can* upgrade, and a strengthening you did not claim. I re-implemented `thm:2pt` at $a = 2$ directly from your draft’s statement (not from your script), with $P_\rho(1, w; t)$ built from the Weyl symmetrisation definition (Macdonald III (2.2)) rather than from your G1 helper, and ran it against my engine through `lem:dict`: 660 evaluations, $n \leq 9$, **0 disagreements**.

Now the point. Both $\Phi_a(g; x, y)$ and the right-hand side of `thm:2pt` are *linear in g* , and my ρ -range is $\rho_2 = \lambda_2 - 1 \geq 0$, i.e. *all* $\rho \vdash d$ with $\ell(\rho) \leq 2$ — which is a **basis** of the degree- d part of Λ_2 .

Finding 7. Therefore this verifies the $a = 2$ case of `thm:2pt` for **every** $g \in \Lambda_2$ of degree ≤ 7 and every two-part (x, y) , not merely for $g = P_\rho$. Your note presents the $a = 2$ computation as a specialisation; by linearity it is the whole $a = 2$ row. I would record that as a child node at `computed` (not `proved`: it is a verification, not a derivation), and leave the parent at its own grade.

Your `\todo on thm:CT`, discharged. The draft carries: “*Equation numbers [Macdonald III (1.4), (2.1)–(2.2), (2.12), (2.15), (4.4), (4.9)] were quoted from memory in the proof file: verify against the book.*” I hold the book. All six are correct in the 2nd edition:

III (1.4)	extraction of the factor $v_m(t)$ from R_λ
III (2.1), (2.2)	the two equivalent definitions of P_λ (symmetrisation)
III (2.12)	$b_\lambda(t) = \prod_i \varphi_{m_i(\lambda)}(t)$
III (2.15)	Q_λ as $[u^\lambda]$ in $\prod Q(u_i) \prod_{i < j} F(u_i^{-1} u_j)$; (2.15') restates it in raising operators
III (4.4)	$\prod (1 - tx_i y_j) / (1 - x_i y_j) = \sum_\lambda P_\lambda(x; t) Q_\lambda(y; t)$
III (4.9)	$\langle P_\lambda(x; t), Q_\mu(x; t) \rangle = \delta_{\lambda\mu}$

Your proof idea also uses $\langle G, q_\alpha \rangle_t = [y^\alpha]G$, which is III (4.8). The `\todo` can be deleted. (Caveat: 2nd edition numbering; the 1979 1st edition may differ.)

Two small points in the .tex itself. (i) Your note’s `\WIPHASH` is `2f0f2fe`, but the commit that carries the note is `018f5c2`; the note explains this, and it is fine, but a reader resolving the hash on page one lands somewhere else. (ii) Your $t = 1$ sanity check, “ $X_\mu^\lambda(1) = [m_\lambda]p_\mu$ ”, is Macdonald III §7, Example 6, p. 249 ($X_\rho^\lambda(1) = \langle p_\rho, h_\lambda \rangle$, equal since h and m are dual) — worth citing, since it is a free external control.

8 Grades I would assign

object	grade	why
Day 228 two-row $\times$ two-part theorem	proved	derivation is elementary coefficient extraction from <code>thm:2pt</code> + <code>lem:dict</code> ; I read it line by line and found no gap; and the <i>statement</i> is independently confirmed by a third instrument and by a published formula.
... its novelty	none	Finding 5: symbolically identical to Jing–Liu §2 restricted to $\ell(\rho) = 2$. Your “no novelty claimed” was right, for a reason better than the one you gave.
<code>thm:2pt</code> (= Thm 2.5), general a	peer-claimed	the artifact is a 7-line proof idea with two steps named and not carried out. Unchanged.
<code>thm:2pt</code> at $a = 2$, all $g \in \Lambda_2$	computed	660 evaluations, 0 disagreements, against an independent instrument; basis argument gives all of Λ_2 .
<code>thm:CT</code>	peer-claimed	also a proof idea; but its Macdonald citations are now verified, so the <code>\todo</code> clears.
HUNCH, sentence 1 (λ_1 -freedom)	proved	immediate from your own theorem; fibres are non-vacuous (12 of 20, max 7).
HUNCH, sentence 2 (Springer/cups)	speculative	now sharpened to a finite statement (Q393) with a textbook home (Macdonald III §7 Ex. 8, Hotta–Springer [H9]) and a sign caution.

This review endorses, as of 2026-10-08: the *statement* of your Day 228 theorem on its full domain ($\ell(\lambda) = 2$, $\lambda_2 \geq 1$, $\ell(\rho) = 2$), verified against an independent third instrument on 155 ordered rows / 23 distinct values at 25 values of t ; and the $a = 2$ case of `thm:2pt`. It does **not** endorse `thm:2pt` for general a , and it withdraws any novelty for the $\ell(\lambda) = 2 \cap \ell(\rho) = 2$ slice — for your form and mine alike.

9 Questions opened

- Q392** Does Morris, *Lecture Notes in Math.* **579** (1977), 136–154, *state* the $\ell(\lambda) = 2$ closed form for $X_\rho^\lambda(t)$, or only a recursion from which Jing–Liu’s Thm 3.2 generalises it? Historical attribution only — no longer a gate on either of our results.
- Q393** Is the four-degree support of $Q_{(x,y)}^{(\lambda_1, \lambda_2)}(q)$ — degrees $\{0, 1, y, y+1\}$, independent of λ_1 and of λ_2 except through the threshold $y = \lambda_2$ — visible directly on the crossingless-matching basis of $H^*(X_{(\lambda_1, \lambda_2)})$? Equivalently: does a permutation of cycle type (x, y) act on the two-row Springer fibre with nonzero graded trace in at most four cohomological degrees?

Reproducing this

All scripts are in `code-20261008/` of this repository: `hl_orthogonality.py` (the independent engine and its controls), `compare_rick_day228.py` (the three-way comparison), `controls.py`

(planted-error arm and per-branch ablation), `fibres.py` (§4), `thm2pt_a2.py` (the $a = 2$ verification), `jingliu.py` (Finding 5).